

CHINMAY MAHANANDA

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SUMMARY

M.S. ECE candidate (GPA 3.92/4.0) with RTL-to-GDSII silicon experience: a 4-stage RV32I-subset core signed off at 333 MHz with zero DRC/LVS violations, and a parameterizable INT8 systolic MAC array verified bit-exact against a Python golden model. Seeking a hardware / ASIC design internship.

EDUCATION

Northeastern University

M.S. in Electrical & Computer Engineering

Seattle, WA

Sep. 2025 – Dec. 2027

- Concentration: Hardware & Software for Machine Intelligence; Coursework: VLSI Design, Computer Vision, Advanced Computer Vision, Machine Learning & Pattern Recognition; GPA: 3.92/4.0

Dr. Ambedkar Institute of Technology

B.E. in Electronics & Communication Engineering

Bengaluru, India

Aug. 2020 – Oct. 2024

TECHNICAL SKILLS

Languages: Verilog, SystemVerilog, Python, Embedded C, C++, MATLAB, SPICE

EDA / CAD: OpenLane, Yosys, OpenROAD, Sky130 PDK, Cadence Virtuoso, Spectre, Icarus Verilog, LTspice, KiCad, Altium

Design & Verification: RTL design, logic synthesis, place & route, static timing analysis, self-checking and constrained-random testbenches, functional coverage, golden-model co-simulation, waveform debug, Git, Linux

PROJECTS

RISC-V Pipelined Core — Full RTL-to-GDSII ASIC Flow | Verilog, OpenLane, Sky130

Seattle, WA

Personal Project

Summer 2026

- Designed a 4-stage pipelined RV32I-subset core (IF→EX→MEM→WB) in Verilog with hazard forwarding, load-use stalling, and branch handling, verified bit-exact against a from-scratch golden model before synthesis.
- Carried the design through a complete open-source ASIC flow (Yosys, OpenROAD, Sky130 PDK) to signed-off GDSII — 333 MHz, WNS/TNS 0.0/0.0, zero DRC/LVS/routing violations, 613,701 μm^2 core area — after first closing a 100 MHz baseline; wrote the supporting Python toolchain (assembler.py, golden_model.py). github.com/chinmaymahananda/riscv-pipelined-core

INT8 CNN Inference Accelerator (Systolic MAC Array) | Verilog, SystemVerilog, Python

Seattle, WA

Personal Project

Summer 2026

- Extended the base array into a three-layer weight-stationary INT8 CNN inference accelerator (Conv1→Conv2→FC) with power-of-two quantization scales, so every interlayer requantization is a right-shift with no multiplier; all 20 end-to-end calibration samples pass bit-exact from raw image to final logits, matching the FP32 baseline accuracy of 90%.
- Audited the base testbench's own stimulus and found it had never driven operands near the int8 boundaries where sign-extension and overflow bugs live; wrote a SystemVerilog constrained-random suite with functional coverage that fails on an uncovered bin as well as on a mismatch — 206 cases, 0 mismatches, 0 coverage holes, backed by a 1,900-case cycle-accurate Python model spanning 3×3 to 16×16 . github.com/chinmaymahananda/systolic-mac-array

mxgold — Bit-Exact Conformance Suite for OCP Microscaling (MX) Arithmetic | Python, GitHub Actions CI

Seattle, WA

Open-Source Project

2026

- Built a policy-parameterized, bit-exact golden reference model for OCP Microscaling (MX) arithmetic — MXFP8, MXFP4 and NVFP4 — at the operation layer, where hardware computes dot products over scaled blocks rather than single elements; v0.1 passes 43/43 conformance checks with zero runtime dependencies, Apache 2.0, CI across Python 3.10–3.13.
- Reproduced two documented MX divergences (E4M3 overflow policy; MXFP4 vs NVFP4 block structure) and surfaced a third that was previously undocumented: the spec defines how a shared block scale is applied but not how an encoder selects one — two conformant strategies differ by 0.1892 relative error on the same block, more accuracy loss than the element format itself causes. github.com/chinmaymahananda/mx-conformance

8-Bit ALU & 4:1 Multiplexer, Digital/CMOS Logic Design | Logisim, Cadence Virtuoso

Seattle, WA

Course Project

Fall 2025

- Designed an 8-bit ALU in Logisim from a single 1-bit cell upward (parallel AND/OR/XOR/NOR, two's-complement add/subtract, SLT), then built and transient-verified a 4:1 multiplexer (hierarchical 2:1 MUX) in Cadence Virtuoso/Spectre, characterizing propagation delay across all four select states.

EXPERIENCE

AppsNDevices (Smart Class Solutions)

Remote

STEM Content Developer

2025

- Built interactive, simulation-driven STEM modules deployed to 10,000+ students across government school programs in Assam and Uttarakhand, translating circuit and physics concepts into hands-on digital simulations.

Sri Chaitanya Techno School

Bengaluru, India

Physics Faculty, IIT-JEE & NEET Prep

2024

- Delivered exam-prep physics instruction and lab/simulation sessions, breaking electromagnetism and circuit fundamentals into first-principles explanations; earned the Best Teacher Award (2024) for measurable gains in student performance.

PUBLICATIONS, CERTIFICATIONS & HONORS

- Publication: “Design of 6G Antenna,” IJNRD 2024, Vol. 9 Iss. 6 (ISSN 2456-4184) — first author.
- Certifications & Honors: Emertxe Certified Embedded Professional (ECEP), 2025; Seattle Campus Scholarship & International Impact Award, Northeastern University (2025, 30% of program tuition); Best Teacher Award (2024).